

Solution Requirements

Date	27 June 2026
Team ID	xxxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Functional Requirements (FR) define the specific behaviors, functions or features the solution must have.
Non-Functional Requirements (NFR) specify the criteria that judge how well a system performs its functions.

Step 1: Functional Requirements (FR)

S.Ns so	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	As this is a public-facing prediction tool, login is not mandatory for general use; an optional admin login should protect model-management actions.	Low
2	Authorization Levels	Define two user roles: General User (submit input, view prediction) and Admin (retrain/update the model, view logs).	Medium
3	External Interfaces	Expose a simple web interface and a /predict endpoint that accepts indicator values and returns the predicted HDI category.	High
4	Transaction Processing	Accept user input from the web form, validate it, run it through the trained model, and return a result within the same request/response cycle.	High
5	Reporting	Display the predicted HDI category clearly on the result page; optionally log past requests for later analysis.	Medium
6	Business Rules	The predicted output must always map to one of four standard categories — Low, Medium, High or Very High Human Development.	High
7	Compliance / Legal / Regulations	Where country-level statistics are used, the indicators and thresholds should align with the publicly published UNDP HDI methodology for credibility.	Medium
8	Other	Provide clear error handling so invalid or missing form input returns a helpful message instead of crashing the app.	Medium

Step 2: Non-Functional Requirements (NFR)

S · N	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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	Performance & Speed	The response time for returning a prediction after form submission.	Under 2 seconds per prediction request
2	Scalability	How the system handles multiple users predicting at the same time.	Supports at least 50 concurrent requests on the hosting platform

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Protection of submitted data and the trained model file.	No personally identifiable data stored; HTTPS enforced in production
4	Reliability & Availability	System uptime and tolerance to bad or malformed input.	99% uptime; graceful error page instead of a server crash
5	Usability & Accessibility	How easy the form and results are to use across devices.	Usable on desktop and mobile; meets basic WCAG 2.1 AA guidelines
6	Other (Maintainability)	Ability to update or retrain the model without changing application code.	model.pkl can be swapped/retrained independently of app.py